

National University of Technology

Machine Learning Lab- CS3504

LAB Submitted to:

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Submitted By:

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Reg. No: **F22607010**
Batch: **Fall AI 2022**
Lab : **02**

Q.NO.1:

```
import pandas as pd

df = pd.read_csv('employee_data.csv')

print(df.sample(5))
```

```
    EmployeeID           Name   Age Department   Salary JoiningDate  \
20          21    Bob Johnson  45.0    Finance  80000.0    30-09-15
4            5  Charlie Black  30.0         HR      NaN    10-02-21
6            7    David Green  50.0    Finance  90000.0    25-02-16
13          14  Jessica White  31.0         IT      NaN    30-01-21
0            1       John Doe  28.0         HR  50000.0    01-06-19

    PerformanceScore
20                 2
4                  4
6                  3
13                 3
0                  4
```

```
df.isnull().sum()
```

✓ 0.0s

```
EmployeeID          0
Name                1
Age                 2
Department          0
Salary              3
JoiningDate         2
PerformanceScore    0
dtype: int64
```

```python
df = df.drop(index=7)
median_age = df['Age'].mean().astype(int)
df['Age'].fillna(median_age, inplace=True)

df.reset_index(drop=True, inplace=True)

df.to_csv('updated_file.csv', index=False)

print(df.sample(5))
```

```python
df = df.drop(index=7)
median_age = df['Age'].mean().astype(int)
df['Age'].fillna(median_age, inplace=True)

df.reset_index(drop=True, inplace=True)

df.to_csv('updated_file.csv', index=False)

print(df.sample(5))
```

✓ 0.0s

```
    EmployeeID           Name   Age Department   Salary JoiningDate  \
1            2     Jane Smith  36.0         IT      NaN    15-03-18
12          14  Jessica White  31.0         IT      NaN    30-01-21
6            7    David Green  50.0    Finance  90000.0    25-02-16
7            9    George Blue  40.0         IT  75000.0    15-07-19
19          21    Bob Johnson  45.0    Finance  80000.0    30-09-15

    PerformanceScore
1                  5
12                 3
6                  3
7                  4
19                 2
```

```python
median_salaries = df.groupby('Department')['Salary'].mean().astype(int)

def fill_missing_salary(row):
    if pd.isna(row['Salary']):
        return median_salaries[row['Department']]
    return row['Salary']


df['Salary'] = df.apply(fill_missing_salary, axis=1)

df.to_csv('updated_file.csv', index=False)

print(df)
```

```
    EmployeeID           Name   Age Department   Salary JoiningDate  \
0            1       John Doe  28.0         HR  50000.0    01-06-19
1            2     Jane Smith  36.0         IT  70666.0    15-03-18
2            3    Alice Brown  36.0         IT  70000.0    20-01-20
3            4    Bob Johnson  45.0    Finance  80000.0    30-09-15
4            5  Charlie Black  30.0         HR  55250.0    10-02-21
5            6    Emily White  38.0         IT  72000.0    12-11-17
6            7    David Green  50.0    Finance  90000.0    25-02-16
7            9    George Blue  40.0         IT  75000.0    15-07-19
8           10     Helen Gray  35.0    Finance  85000.0    20-08-18
9           11      Chris Red  27.0         IT  62000.0    15-09-20
10          12   Karen Yellow  42.0         HR  58000.0         NaN
11          13    Tommy Black  33.0    Finance  72000.0    10-03-19
12          14  Jessica White  31.0         IT  70666.0    30-01-21
13          15      Nina Gray  26.0         HR  49000.0    22-04-22
14          16    Oscar Brown  47.0    Finance  88000.0    12-12-18
15          17    Susan Green  29.0         IT  71000.0         NaN
16          18       Max Blue  36.0    Finance  77000.0    05-11-20
17          19      Linda Red  34.0         HR  64000.0    25-07-19
18          20    Kevin Black  39.0         IT  74000.0    05-03-21
19          21    Bob Johnson  45.0    Finance  80000.0    30-09-15
```

```python
most_frequent_date = df['JoiningDate'].mode()[0]

df['JoiningDate'].fillna(most_frequent_date, inplace=True)
```

```
df.to_csv('updated_emoployees_data.csv', index=False)

print(df)
```

```
    EmployeeID           Name   Age Department   Salary JoiningDate  \
0            1       John Doe  28.0         HR  50000.0    01-06-19
1            2     Jane Smith  36.0         IT  70666.0    15-03-18
2            3    Alice Brown  36.0         IT  70000.0    20-01-20
3            4    Bob Johnson  45.0    Finance  80000.0    30-09-15
4            5  Charlie Black  30.0         HR  55250.0    10-02-21
5            6    Emily White  38.0         IT  72000.0    12-11-17
6            7    David Green  50.0    Finance  90000.0    25-02-16
7            9    George Blue  40.0         IT  75000.0    15-07-19
8           10     Helen Gray  35.0    Finance  85000.0    20-08-18
9           11      Chris Red  27.0         IT  62000.0    15-09-20
10          12   Karen Yellow  42.0         HR  58000.0    30-09-15
11          13    Tommy Black  33.0    Finance  72000.0    10-03-19
12          14  Jessica White  31.0         IT  70666.0    30-01-21
13          15      Nina Gray  26.0         HR  49000.0    22-04-22
14          16    Oscar Brown  47.0    Finance  88000.0    12-12-18
15          17    Susan Green  29.0         IT  71000.0    30-09-15
16          18       Max Blue  36.0    Finance  77000.0    05-11-20
17          19      Linda Red  34.0         HR  64000.0    25-07-19
18          20    Kevin Black  39.0         IT  74000.0    05-03-21
19          21    Bob Johnson  45.0    Finance  80000.0    30-09-15
```

```
duplicate_rows = df[df.duplicated()]


duplicate_columns = df.columns[df.columns.duplicated()]


if not duplicate_rows.empty:
    print("Duplicate Rows:")
    print(duplicate_rows)
else:
    print("No duplicate rows found.")
```

```python
if len(duplicate_columns) > 0:
    print("Duplicate Columns:")
    print(duplicate_columns)
else:
    print("No duplicate columns found.")
```

```
No duplicate rows found.
No duplicate columns found.
```

```python
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()

df[['Salary', 'PerformanceScore']] = scaler.fit_transform(df[['Salary',
'PerformanceScore']])

df.to_csv('updated_file.csv', index=False)

print(df)
```

```
    EmployeeID           Name   Age Department    Salary JoiningDate  \
0            1       John Doe  28.0         HR -1.834170    01-06-19
1            2     Jane Smith  36.0         IT -0.001162    15-03-18
2            3    Alice Brown  36.0         IT -0.060234    20-01-20
3            4    Bob Johnson  45.0    Finance  0.826734    30-09-15
4            5  Charlie Black  30.0         HR -1.368512    10-02-21
5            6    Emily White  38.0         IT  0.117160    12-11-17
6            7    David Green  50.0    Finance  1.713702    25-02-16
7            9    George Blue  40.0         IT  0.383250    15-07-19
8           10     Helen Gray  35.0    Finance  1.270218    20-08-18
9           11      Chris Red  27.0         IT -0.769808    15-09-20
10          12   Karen Yellow  42.0         HR -1.124595    30-09-15
11          13    Tommy Black  33.0    Finance  0.117160    10-03-19
12          14  Jessica White  31.0         IT -0.001162    30-01-21
13          15      Nina Gray  26.0         HR -1.922867    22-04-22
14          16    Oscar Brown  47.0    Finance  1.536308    12-12-18
15          17    Susan Green  29.0         IT  0.028463    30-09-15
16          18       Max Blue  36.0    Finance  0.560644    05-11-20
17          19      Linda Red  34.0         HR -0.592415    25-07-19
18          20    Kevin Black  39.0         IT  0.294553    05-03-21
19          21    Bob Johnson  45.0    Finance  0.826734    30-09-15
```

```python
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()


df[['Salary', 'PerformanceScore']] = scaler.fit_transform(df[['Salary',
'PerformanceScore']])


df.to_csv('updated_file.csv', index=False)

print(df)
```

```
    EmployeeID           Name   Age Department    Salary JoiningDate  \
0            1       John Doe  28.0         HR  0.024390    01-06-19
1            2     Jane Smith  36.0         IT  0.528439    15-03-18
2            3    Alice Brown  36.0         IT  0.512195    20-01-20
3            4    Bob Johnson  45.0    Finance  0.756098    30-09-15
4            5  Charlie Black  30.0         HR  0.152439    10-02-21
5            6    Emily White  38.0         IT  0.560976    12-11-17
6            7    David Green  50.0    Finance  1.000000    25-02-16
7            9    George Blue  40.0         IT  0.634146    15-07-19
8           10     Helen Gray  35.0    Finance  0.878049    20-08-18
9           11      Chris Red  27.0         IT  0.317073    15-09-20
10          12   Karen Yellow  42.0         HR  0.219512    30-09-15
11          13    Tommy Black  33.0    Finance  0.560976    10-03-19
12          14  Jessica White  31.0         IT  0.528439    30-01-21
13          15      Nina Gray  26.0         HR  0.000000    22-04-22
14          16    Oscar Brown  47.0    Finance  0.951220    12-12-18
15          17    Susan Green  29.0         IT  0.536585    30-09-15
16          18       Max Blue  36.0    Finance  0.682927    05-11-20
17          19      Linda Red  34.0         HR  0.365854    25-07-19
18          20    Kevin Black  39.0         IT  0.609756    05-03-21
19          21    Bob Johnson  45.0    Finance  0.756098    30-09-15
```

```python
df_encoded = pd.get_dummies(df, columns=['Department'])

df_encoded.to_csv('updated_file.csv', index=False)

print(df_encoded)
```

```
    Department_Finance  Department_HR  Department_IT
0                False           True          False
1                False          False           True
...
16                True          False          False
17               False           True          False
18               False          False           True
19                True          False          False
```

```python
from datetime import datetime


current_year = datetime.now().year


df['YearsAtCompany'] = current_year - pd.to_datetime(df['JoiningDate'],
format='%d-%m-%y').dt.year


df.to_csv('updated_file.csv', index=False)


print(df)
```

```
19          21    Bob Johnson  45.0    Finance  0.736098    30-

    PerformanceScore  YearsAtCompany
0           0.666667               5
1           1.000000               6
...
16          0.333333               4
17          1.000000               5
18          0.666667               3
19          0.000000               9
```